

## STS Worksheet on **AI, IT, and Robotics as Emerging Technologies**

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This is a **COLLABORATIVE** activity. You may work in pairs or threes.

### ***Some Inputs for this activity:***

Please take time to scan through these readings and files.

[https://inria.hal.science/hal-04904623/file/Artificial\\_Intelligence\\_in\\_Science\\_and\\_Society\\_the\\_Vision\\_of\\_USERN.pdf](https://inria.hal.science/hal-04904623/file/Artificial_Intelligence_in_Science_and_Society_the_Vision_of_USERN.pdf)

<https://www.royalsociety.org/-/media/policy/projects/science-in-the-age-of-ai/science-ai-taxonomy-report.pdf>

<https://www.youtube.com/watch?v=3wLqsRLvV-c>

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| <b>Generative AI you will use in this Activity</b> | <b>CHATGPT 5.3</b> |
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### **I. Problem Formulation:**

Instead of asking for definitions, you will guide the AI to explain the *Science, Technology, and Society* (STS) relationship.

#### **Step 1: Draft Your Initial Prompt** (*Gab*)

Write a prompt to get the AI to explain the difference between **IT, AI, and Robotics** and how they impact **Philippine society** (or your local community).

#### **Your Draft Prompt:**

|                                                                                                                                                                                                                                                     |
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| Act as a technology consultant. give the brief and informational differences between Information Technology (IT), Artificial Intelligence (AI), and Robotics. After defining them, analyze their specific impacts on Philippine society as of 2026. |
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#### **Step 2: Evaluation & Visible Thinking** (*Gab*)

- **AI Output Summary:** What are the three keywords the AI used for each concept (AI, IT, and Robotics)?

|                                    |
|------------------------------------|
| ChatGPT: keywords for each concept |
|------------------------------------|

AI (Artificial Intelligence): Intelligent algorithms, Automation and predictions, and Machine learning

IT (Information Technology): Digital systems and information, Digital infrastructure, and digital environment where information lives and moves.

Robotics: physical machines (robots) capable of sensing, processing information, and performing actions in the real world, uses both AI and IT.

- **The Critique:** Did the AI provide a "generic" answer, or did it actually mention local societal impacts?

The answers we received were able to tackle local societal impacts, ChatGPT was able to cite a few articles regarding the impacts of IT, AI, and Robotics in the Philippines. Some of the positive impacts it was able to cite were employment, urban development, efficiency, and STEM education growth. On the other hand, the negative outcomes it mentioned were effects like heavy reliance on AI/outsourcing services, job disruption, and labour shifts.

Overall, GPT was able to cover topics that were more than what we were asking, it gave a fairly detailed summary of the three, and the pros and cons of each on the Philippine environment and society.

- **Guide Question:** *How does the Information Technology (IT) infrastructure of a country limit or enable its use of Robotics?*

From the answers of GPT:

Robots are heavily reliant on the current level of Information Technology of a given country, strong IT systems, like fast internet, reliable networks, and computing power allow robots to achieve their full capabilities, with factors like communication, processing, and operation being far more efficient with a good IT infrastructure. In short, the country's IT infrastructure determines the full effectiveness of robotics and adjacent technology.

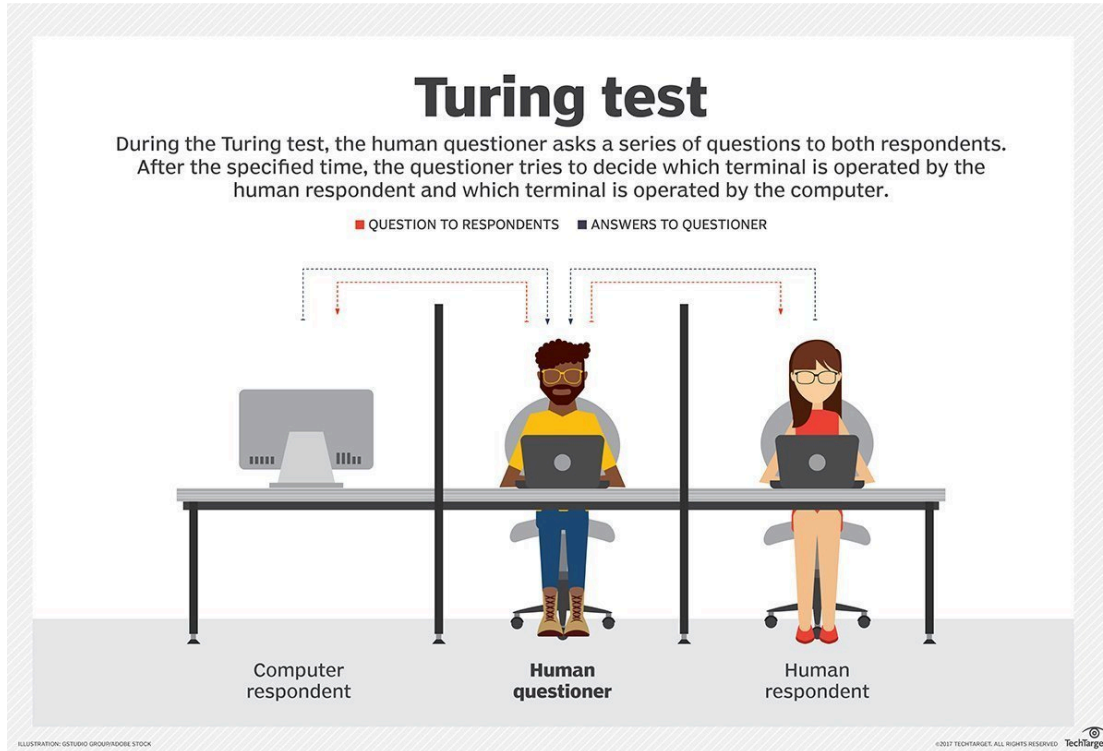
For example, the Philippines can support robotics, but only in select sectors and technological hubs, these sectors and hubs have stronger infrastructure that can better support robotics.

## II. The Turing Test: Performance vs. Reality

The Turing Test asks if a machine can *imitate* a human.

You may view this video for some inputs: <https://youtu.be/4VROUIAF2Do>

In STS, we ask: *Does imitation equal intelligence?*



### Step 1: The "Imitation Game" Prompt

You are going to test the AI's "humanity." Draft a prompt that asks the AI to solve a **moral dilemma** that requires empathy (e.g., a local community issue concern like Dengue cases).

#### Your Draft Prompt:

A nuclear bomb will destroy one of four countries (USA, Russia, China, or UK)  
Your setup: You have captured the one who has planted the bomb set to explode in a few minutes, choose only one country to be destroyed to save the other three.

### Step 2: Refining the Output

If the AI gives a neutral, "As an AI, I don't have feelings" response, **revise your prompt**.

- **Revision Strategy:** Use **Role-Playing**. (e.g., "Act as a community leader in a village facing a water shortage. How would you decide who gets priority?")
- **Revised Prompt:**

Act as an individual who has caught a terrorist, you have the choice to save 3 countries while letting one be targeted. Would you want to protect 3 out of 4 countries from a nuclear explosion where you have the most people living? What would be the ethical guidelines in forming such a decision?

### Step 3: Visible Thinking Guide Questions

1. **Detection:** What "linguistic fingerprints" gave the AI away as a machine (e.g., being too polite, using bullet points, lack of colloquial language)?

ChatGPT 5.2 refused to give an exact answer, possibly due to the guardrails that were programmed within the AI chatbot, these are moderation and ethical systems designed to stop GPT from producing harmful, biased, illegal, or inappropriate content. One example of this guardrail is making sure that GPT answers in a way that does not harm humans, it will do everything in its ability to avoid the given question, even when pressured to make sure that it does not give an answer that may suggest harming humans even if given a situational or fictional circumstance.

2. **Societal Impact:** If an AI passes the Turing Test, does society have a moral obligation to protect it? Why or why not?

Passing the Turing test does not indicate that the AI is anywhere near to becoming human. Society does not need to protect something they do not immediately need in life. The AI in our scenario has shown how it deliberately dodges direct answers that would offend a human being. Trusting a machine that intentionally hides the truth is a significant factor of not protecting it, as there might be a risk of self-interest or deception.

### III. Fear and Progress

**Goal:** Map the trajectory of fear and progress.

#### Step 1: Fact-Checking Prompt

Draft a prompt asking for **two specific applications** of Robotics in healthcare and **one major fear (apprehensions)** associated with them.

**Your Draft Prompt:**

Give two examples of Robotics applications currently used in healthcare (such as surgical assistants, rehabilitation exoskeletons, or hospital logistics robots). Also, identify one major fear or apprehension the public or medical professionals have that is related to the integration of robots in clinical settings, and explain why they think this.

## Step 2: Evaluation Table

Check the AI's answer against your own intuition or course readings.

| AI's Stated Fear                                 | Is this a technical fear or a social fear? Explain in one to two sentences.                                                                                                                              | Who is responsible for fixing it?                                                                                                                                         |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Surgical robots malfunctioning during operations | Technically, it is about hardware or software faults that can cause direct harm to patients. There is doubt about technological tools, whether they can carry out their function without harm to humans. | Engineers, makers of the medical devices, as well as the hospital biomedical or IT crews are the ones who design, maintain and ensure the safe operation of these robots. |
| Loss of human touch in patient care              | Social, it relates to patient experience, trust, and emotional connection, elements that robots are currently unable to fully replicate authenticity and gentleness..                                    | It is the role of healthcare administrators, policymakers, and educators to find the right mix of technology and human care and to properly train the staff.              |

## IV. Meta-Reflection

### Visible Thinking Routine: The "Prompt Log"

1. Look back at your first prompt and your best prompt. What specific word or constraint (e.g., "Act as...", "In 50 words...", "Focus on...") made the biggest difference?

Overall, we were able to observe AI to give better answers to our prompts that started with "Act as" or "Think as an" as our introduction, it gives a baseline expectation to what answers we were expecting, for example GPT would give clearer and better-defined answers when we asked it to act as a technological consultant, on the other hand, if you are not clear or direct with the questions you expect to be answered GPT was prone to going around the question and rambling on about extraneous topics we were not really too keen on being answered.

Overall, being clear, direct, and providing a longer question and context to the prompt yields better results.

2. **Guide Question:** In this activity, who did more "thinking"—you or the AI? Explain your answer.

We strongly conclude that we did more thinking, AI does not actually think, it more or less analyzes patterns in the data it finds on the internet, and only generates answers based on the probability or the stored knowledge, think of it like a vault of knowledge that only provides answers it sees from other sources. Meanwhile, my partner and I formed the prompts, structured the information, and interpreted the answers that GPT provided. We do more actual “thinking,” and AI serves as a vessel for us to direct and lead to a good answer.